Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.30_e1)

Metric

$$ds^2 = 2\left[(K+1)^2 v^2 + lv + H\right] \mathrm{d}u \otimes \mathrm{d}u - \frac{2\left[b_x(u,x,y)^2 + b_y(u,x,y)^2\right]}{(K+1)^2\left(x^2+y^2+1\right)^2} \mathrm{d}u \otimes \mathrm{d}u - \mathrm{d}u \otimes \mathrm{d}v - \mathrm{d}v \otimes \mathrm{d}u + \frac{2\, b_x(u,x,y)}{(K+1)^2\left(x^2+y^2+1\right)} \mathrm{d}u \otimes \mathrm{d}x + \frac{2\, b_x(u,x,y)}{(K+1)^2\left(x^2+y^2+1\right)} \mathrm{d}x \otimes \mathrm{d}u + \frac{2\, b_y(u,x,y)}{(K+1)^2\left(x^2+y^2+1\right)} \mathrm{d}u \otimes \mathrm{d}y + \frac{2\, b_y(u,x,y)}{(K+1)^2\left(x^2+y^2+1\right)} \mathrm{d}y \otimes \mathrm{d}u + \frac{2}{(K+1)^2\left(x^2+y^2+1\right)} \mathrm{d}x \otimes \mathrm{d}x + \frac{2}{(K+1)^2\left(x^2+y^2+1\right)} \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$(x^\mu) = (u, v, x, y)$

Metric Functions

$K(u) = K$

$H(u,x,y) = H$

Null Tetrad

Covariant components

$l = (1)\ du$

$$n = \left(-v^2K^2 - 2\,v^2K - lv - v^2 + \frac{b_x\left(u,x,y\right)^2}{(x^4+2\,x^2y^2+y^4+2\,x^2+2\,y^2+1)(K^2+2\,K+1)} + \frac{b_y\left(u,x,y\right)^2}{(x^4+2\,x^2y^2+y^4+2\,x^2+2\,y^2+1)(K^2+2\,K+1)} - H\right) du + (1)\ dv + \left(-\frac{2\,b_x\left(u,x,y\right)}{(x^2+y^2+1)(K^2+2\,K+1)}\right) dx + \left(-\frac{2\,b_y\left(u,x,y\right)}{(x^2+y^2+1)(K^2+2\,K+1)}\right) dy$$

$$m = \left(\frac{1}{\sqrt{x^2+y^2+1}(K+1)}\right) dx + \left(\frac{i}{\sqrt{x^2+y^2+1}(K+1)}\right) dy$$

$$\overline{m} = \left(\frac{1}{\sqrt{x^2+y^2+1}(K+1)}\right) dx + \left(-\frac{i}{\sqrt{x^2+y^2+1}(K+1)}\right) dy$$

Contravariant components

$$l = (-1)\ dv$$

$$\begin{aligned} n = &-\mathrm{d}u + \left(-\frac{v^2x^4K^4}{D_1} - \frac{2\,v^2x^2y^2K^4}{D_1} - \frac{v^2y^4K^4}{D_1} - \frac{4\,v^2x^4K^3}{D_1} - \frac{8\,v^2x^2y^2K^3}{D_1} - \frac{4\,v^2y^4K^3}{D_1} - \frac{lvx^4K^2}{D_1} - \frac{6\,v^2x^4K^2}{D_1} \right. \\ &\left. - \frac{2\,lvx^2y^2K^2}{D_1} - \frac{12\,v^2x^2y^2K^2}{D_1} - \frac{lvy^4K^2}{D_1} - \frac{6\,v^2y^4K^2}{D_1} - \frac{2\,v^2x^2K^4}{D_1} - \frac{2\,v^2y^2K^4}{D_1} - \frac{2\,lvx^4K}{D_1} - \frac{4\,v^2x^4K}{D_1} - \frac{4\,lvx^2y^2K}{D_1} + \cdots \right) \end{aligned}$$

$$D_1 = x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1$$

$$\begin{aligned} m = &\left(\frac{\sqrt{x^2+y^2+1}b_x\,(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1} + \frac{i\,\sqrt{x^2+y^2+1}b_y\,(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1}\right) dv \\ &+ \left(\frac{x^2K}{2\,\sqrt{x^2+y^2+1}} + \frac{y^2K}{2\,\sqrt{x^2+y^2+1}} + \frac{x^2}{2\,\sqrt{x^2+y^2+1}} + \frac{y^2}{2\,\sqrt{x^2+y^2+1}} + \frac{K}{2\,\sqrt{x^2+y^2+1}} + \frac{1}{2\,\sqrt{x^2+y^2+1}}\right) dx \\ &+ \left(\frac{i\,x^2K}{2\,\sqrt{x^2+y^2+1}} + \frac{i\,y^2K}{2\,\sqrt{x^2+y^2+1}} + \frac{i\,x^2}{2\,\sqrt{x^2+y^2+1}} + \frac{i\,y^2}{2\,\sqrt{x^2+y^2+1}} + \frac{i\,K}{2\,\sqrt{x^2+y^2+1}} + \frac{i}{2\,\sqrt{x^2+y^2+1}}\right) dy \end{aligned}$$

$$\begin{aligned} \overline{m} = &\left(\frac{\sqrt{x^2+y^2+1}b_x\,(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1} - \frac{i\,\sqrt{x^2+y^2+1}b_y\,(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1}\right) dv \\ &+ \left(\frac{x^2K}{2\,\sqrt{x^2+y^2+1}} + \frac{y^2K}{2\,\sqrt{x^2+y^2+1}} + \frac{x^2}{2\,\sqrt{x^2+y^2+1}} + \frac{y^2}{2\,\sqrt{x^2+y^2+1}} + \frac{K}{2\,\sqrt{x^2+y^2+1}} + \frac{1}{2\,\sqrt{x^2+y^2+1}}\right) dx \\ &+ \left(-\frac{i\,x^2K}{2\,\sqrt{x^2+y^2+1}} - \frac{i\,y^2K}{2\,\sqrt{x^2+y^2+1}} - \frac{i\,x^2}{2\,\sqrt{x^2+y^2+1}} - \frac{i\,y^2}{2\,\sqrt{x^2+y^2+1}} - \frac{i\,K}{2\,\sqrt{x^2+y^2+1}} - \frac{i}{2\,\sqrt{x^2+y^2+1}}\right) dy \end{aligned}$$

Directional Derivatives

$D = -\partial_v$

$$
\begin{aligned}
\Delta = -\partial_u &- \frac{v^2 x^4 K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,v^2x^2y^2K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{v^2y^4K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{4\,v^2x^4K^3}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{8\,v^2x^2y^2K^3}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{4\,v^2y^4K^3}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{lvx^4K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{6\,v^2x^4K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,lvx^2y^2K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{12\,v^2x^2y^2K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{lvy^4K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{6\,v^2y^4K^2}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,v^2x^2K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,v^2y^2K^4}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{2\,lvx^4K}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{4\,v^2x^4K}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} \\
&- \frac{4\,lvx^2y^2K}{x^4K^2 + 2\,x^2y^2K^2 + y^4K^2 + 2\,x^4K + 4\,x^2y^2K + 2\,y^4K + x^4 + 2\,x^2y^2 + y^4 + 2\,x^2K^2 + 2\,y^2K^2 + 4\,x^2K + 4\,y^2K + 2\,x^2 + 2\,y^2 + K^2 + 2\,K + 1} - \cdots
\end{aligned}
$$

$$\delta = \frac{\sqrt{x^2+y^2+1}b_x(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1} + \frac{i\sqrt{x^2+y^2+1}b_y(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1}\partial_v + \frac{x^2K}{2\sqrt{x^2+y^2+1}} + \frac{y^2K}{2\sqrt{x^2+y^2+1}} + \frac{x^2}{2\sqrt{x^2+y^2+1}} + \frac{y^2}{2\sqrt{x^2+y^2+1}}$$
$$+ \frac{K}{2\sqrt{x^2+y^2+1}} + \frac{1}{2\sqrt{x^2+y^2+1}}\partial_x + \frac{i\,x^2K}{2\sqrt{x^2+y^2+1}} + \frac{i\,y^2K}{2\sqrt{x^2+y^2+1}} + \frac{i\,x^2}{2\sqrt{x^2+y^2+1}} + \frac{i\,y^2}{2\sqrt{x^2+y^2+1}} + \frac{i\,K}{2\sqrt{x^2+y^2+1}} + \frac{i}{2\sqrt{x^2+y^2+1}}\partial_y$$

$$\overline{\delta} = \frac{\sqrt{x^2+y^2+1}b_x(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1} - \frac{i\sqrt{x^2+y^2+1}b_y(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1}\partial_v + \frac{x^2K}{2\sqrt{x^2+y^2+1}} + \frac{y^2K}{2\sqrt{x^2+y^2+1}} + \frac{x^2}{2\sqrt{x^2+y^2+1}} + \frac{y^2}{2\sqrt{x^2+y^2+1}}$$
$$+ \frac{K}{2\sqrt{x^2+y^2+1}} + \frac{1}{2\sqrt{x^2+y^2+1}}\partial_x - \frac{i\,x^2K}{2\sqrt{x^2+y^2+1}} - \frac{i\,y^2K}{2\sqrt{x^2+y^2+1}} - \frac{i\,x^2}{2\sqrt{x^2+y^2+1}} - \frac{i\,y^2}{2\sqrt{x^2+y^2+1}} - \frac{i\,K}{2\sqrt{x^2+y^2+1}} - \frac{i}{2\sqrt{x^2+y^2+1}}\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{i\,x^2K\frac{\partial}{\partial y}b_x(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)} - \frac{i\,y^2K\frac{\partial}{\partial y}b_x(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)} + \frac{i\,x^2K\frac{\partial}{\partial x}b_y(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)}$$
$$+ \frac{i\,y^2K\frac{\partial}{\partial x}b_y(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)} + \frac{i\,yKb_x(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1} - \frac{i\,xKb_y(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1} + \frac{x^2\frac{\partial}{\partial u}K}{x^2K+y^2K+x^2+y^2+K+1}$$
$$+ \frac{y^2\frac{\partial}{\partial u}K}{x^2K+y^2K+x^2+y^2+K+1} - \frac{i\,x^2\frac{\partial}{\partial y}b_x(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)} - \frac{i\,y^2\frac{\partial}{\partial y}b_x(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)} + \frac{i\,x^2\frac{\partial}{\partial x}b_y(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)}$$
$$+ \frac{i\,y^2\frac{\partial}{\partial x}b_y(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)} + \frac{i\,yb_x(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1} - \frac{i\,xb_y(u,x,y)}{x^2K+y^2K+x^2+y^2+K+1} - \frac{i\,K\frac{\partial}{\partial y}b_x(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)}$$
$$+ \frac{i\,K\frac{\partial}{\partial x}b_y(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)} + \frac{\frac{\partial}{\partial u}K}{x^2K+y^2K+x^2+y^2+K+1} - \frac{i\,\frac{\partial}{\partial y}b_x(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)} + \frac{i\,\frac{\partial}{\partial x}b_y(u,x,y)}{2(x^2K+y^2K+x^2+y^2+K+1)}$$

$$\lambda = 0$$

$$\nu = -\frac{\sqrt{x^2+y^2+1}\,x^6K^3\,\frac{\partial}{\partial x}H}{2\,D_2} - \frac{3\,\sqrt{x^2+y^2+1}\,x^4y^2K^3\,\frac{\partial}{\partial x}H}{2\,D_2} - \frac{3\,\sqrt{x^2+y^2+1}\,x^2y^4K^3\,\frac{\partial}{\partial x}H}{2\,D_2} - \frac{\sqrt{x^2+y^2+1}\,y^6K^3\,\frac{\partial}{\partial x}H}{2\,D_2}$$
$$+ \frac{i\,\sqrt{x^2+y^2+1}\,x^6K^3\,\frac{\partial}{\partial y}H}{2\,D_2} + \frac{3i\,\sqrt{x^2+y^2+1}\,x^4y^2K^3\,\frac{\partial}{\partial y}H}{2\,D_2} + \frac{3i\,\sqrt{x^2+y^2+1}\,x^2y^4K^3\,\frac{\partial}{\partial y}H}{2\,D_2} + \frac{i\,\sqrt{x^2+y^2+1}\,y^6K^3\,\frac{\partial}{\partial y}H}{2\,D_2} + \cdots$$

$$D_2 = x^6K^2 + 3\,x^4y^2K^2 + 3\,x^2y^4K^2 + y^6K^2 + 2\,x^6K + 6\,x^4y^2K + 6\,x^2y^4K + 2\,y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K^2 + 6\,x^2y^2K^2$$
$$+ 3\,y^4K^2 + 6\,x^4K + 12\,x^2y^2K + 6\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K^2 + 3\,y^2K^2 + 6\,x^2K + 6\,y^2K + 3\,x^2 + 3\,y^2 + K^2 + 2\,K + 1$$

$$\alpha = \frac{xK}{4\,\sqrt{x^2+y^2+1}} - \frac{i\,yK}{4\,\sqrt{x^2+y^2+1}} + \frac{x}{4\,\sqrt{x^2+y^2+1}} - \frac{i\,y}{4\,\sqrt{x^2+y^2+1}}$$

$$\beta = -\frac{xK}{4\,\sqrt{x^2+y^2+1}} - \frac{i\,yK}{4\,\sqrt{x^2+y^2+1}} - \frac{x}{4\,\sqrt{x^2+y^2+1}} - \frac{i\,y}{4\,\sqrt{x^2+y^2+1}}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{vx^2K^2}{x^2+y^2+1} + \frac{vy^2K^2}{x^2+y^2+1} + \frac{2\,vx^2K}{x^2+y^2+1} + \frac{2\,vy^2K}{x^2+y^2+1} + \frac{lx^2}{2\,(x^2+y^2+1)} + \frac{vx^2}{x^2+y^2+1} + \frac{ly^2}{2\,(x^2+y^2+1)} + \frac{vy^2}{x^2+y^2+1} + \frac{vK^2}{x^2+y^2+1} - \frac{i\,x^2\,\frac{\partial}{\partial y}b_x\,(u,x,y)}{4\,(x^2+y^2+1)} - \frac{i\,y^2\,\frac{\partial}{\partial y}b_x\,(u,x,y)}{4\,(x^2+y^2+1)}$$
$$+ \frac{i\,x^2\,\frac{\partial}{\partial x}b_y\,(u,x,y)}{4\,(x^2+y^2+1)} + \frac{i\,y^2\,\frac{\partial}{\partial x}b_y\,(u,x,y)}{4\,(x^2+y^2+1)} + \frac{2\,vK}{x^2+y^2+1} + \frac{i\,yb_x\,(u,x,y)}{2\,(x^2+y^2+1)} - \frac{i\,xb_y\,(u,x,y)}{2\,(x^2+y^2+1)} + \frac{l}{2\,(x^2+y^2+1)} + \frac{v}{x^2+y^2+1} - \frac{i\,\frac{\partial}{\partial y}b_x\,(u,x,y)}{4\,(x^2+y^2+1)} + \frac{i\,\frac{\partial}{\partial x}b_y\,(u,x,y)}{4\,(x^2+y^2+1)}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{x^2K^2}{2\left(x^2+y^2+1\right)} - \frac{y^2K^2}{2\left(x^2+y^2+1\right)} - \frac{x^2K}{x^2+y^2+1} - \frac{y^2K}{x^2+y^2+1} - \frac{x^2}{2\left(x^2+y^2+1\right)} - \frac{y^2}{2\left(x^2+y^2+1\right)} - \frac{K^2}{4\left(x^2+y^2+1\right)} - \frac{K}{2\left(x^2+y^2+1\right)} - \frac{1}{4\left(x^2+y^2+1\right)}$$

$$\begin{aligned}
\Phi_{12} = &-\frac{i\,x^4K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x^2y^2K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,y^4K\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^4K\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^2y^2K\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{y^4K\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
&+ \frac{i\,x^4K\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,x^2y^2K\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,y^4K\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^4K\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^2y^2K\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{y^4K\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
&+ \frac{i\,x^2yK\frac{\partial}{\partial x}b_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,y^3K\frac{\partial}{\partial x}b_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x^4\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x^2y^2\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,y^4\frac{\partial^2}{\partial x\partial y}b_x\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^2yK\frac{\partial}{\partial y}b_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
&- \frac{y^3K\frac{\partial}{\partial y}b_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^4\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^2y^2\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{y^4\frac{\partial^2}{(\partial y)^2}b_x\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x^3K\frac{\partial}{\partial x}b_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,xy^2K\frac{\partial}{\partial x}b_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,x^4\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
&+ \frac{i\,x^2y^2\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,y^4\frac{\partial^2}{(\partial x)^2}b_y\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^4\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^2y^2\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{y^4\frac{\partial^2}{\partial x\partial y}b_y\left(u,x,y\right)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^3K\frac{\partial}{\partial y}b_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
&+ \frac{xy^2K\frac{\partial}{\partial y}b_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{3\,x^2Kb_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,xyKb_x\left(u,x,y\right)}{2\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{5\,y^2Kb_x\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{5i\,x^2Kb_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{xyKb_y\left(u,x,y\right)}{2\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{3i\,y^2Kb_y\left(u,x,y\right)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \ldots
\end{aligned}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{i\,x^4 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{i\,x^2 y^2 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{i\,y^4 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{x^4 K \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{x^2 y^2 K \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{y^4 K \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}}$$
$$- \frac{i\,x^4 K \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{i\,x^2 y^2 K \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{i\,y^4 K \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{x^4 K \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{x^2 y^2 K \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{y^4 K \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}}$$
$$- \frac{i\,x^2 y K \frac{\partial}{\partial x} b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{i\,y^3 K \frac{\partial}{\partial x} b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{i\,x^4 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{i\,x^2 y^2 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{i\,y^4 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{x^2 y K \frac{\partial}{\partial y} b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}}$$
$$- \frac{y^3 K \frac{\partial}{\partial y} b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{x^4 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{x^2 y^2 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{y^4 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{i\,x^3 K \frac{\partial}{\partial x} b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{i\,x y^2 K \frac{\partial}{\partial x} b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{i\,x^4 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}}$$
$$- \frac{i\,x^2 y^2 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{i\,y^4 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{x^4 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{x^2 y^2 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{y^4 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{x^3 K \frac{\partial}{\partial y} b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}}$$
$$+ \frac{x y^2 K \frac{\partial}{\partial y} b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{3\,x^2 K b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{i\,x y K b_x(u,x,y)}{2\,(x^2+y^2+1)^{\frac{3}{2}}} + \frac{5\,y^2 K b_x(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{5i\,x^2 K b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{x y K b_y(u,x,y)}{2\,(x^2+y^2+1)^{\frac{3}{2}}} - \frac{3i\,y^2 K b_y(u,x,y)}{4\,(x^2+y^2+1)^{\frac{3}{2}}} - \cdots$$

$$\Phi_{22} = -\frac{x^8 K^4 \frac{\partial^2}{(\partial x)^2} H}{4\,D_2} - \frac{x^6 y^2 K^4 \frac{\partial^2}{(\partial x)^2} H}{D_2} - \frac{3\,x^4 y^4 K^4 \frac{\partial^2}{(\partial x)^2} H}{2\,D_2} - \frac{x^2 y^6 K^4 \frac{\partial^2}{(\partial x)^2} H}{D_2} - \frac{y^8 K^4 \frac{\partial^2}{(\partial x)^2} H}{4\,D_2} - \frac{x^8 K^4 \frac{\partial^2}{(\partial y)^2} H}{4\,D_2} - \frac{x^6 y^2 K^4 \frac{\partial^2}{(\partial y)^2} H}{D_2} - \frac{3\,x^4 y^4 K^4 \frac{\partial^2}{(\partial y)^2} H}{2\,D_2} - \frac{x^2 y^6 K^4 \frac{\partial^2}{(\partial y)^2} H}{D_2} + \cdots$$

$$\Lambda = \frac{x^2 K^2}{6\,(x^2+y^2+1)} + \frac{y^2 K^2}{6\,(x^2+y^2+1)} + \frac{x^2 K}{3\,(x^2+y^2+1)} + \frac{y^2 K}{3\,(x^2+y^2+1)} + \frac{x^2}{6\,(x^2+y^2+1)} + \frac{y^2}{6\,(x^2+y^2+1)} + \frac{K^2}{4\,(x^2+y^2+1)} + \frac{K}{2\,(x^2+y^2+1)} + \frac{1}{4\,(x^2+y^2+1)}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{x^2 K^2}{3\,(x^2+y^2+1)} - \frac{y^2 K^2}{3\,(x^2+y^2+1)} - \frac{2\,x^2 K}{3\,(x^2+y^2+1)} - \frac{2\,y^2 K}{3\,(x^2+y^2+1)} - \frac{x^2}{3\,(x^2+y^2+1)} - \frac{y^2}{3\,(x^2+y^2+1)} - \frac{K^2}{2\,(x^2+y^2+1)} - \frac{K}{x^2+y^2+1} - \frac{1}{2\,(x^2+y^2+1)}$$

$$
\begin{aligned}
\Psi_3 = & -\frac{i\,x^4 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x^2 y^2 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,y^4 K \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^4 K \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^2 y^2 K \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{y^4 K \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
& + \frac{i\,x^4 K \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,x^2 y^2 K \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,y^4 K \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^4 K \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^2 y^2 K \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{y^4 K \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
& + \frac{i\,x^2 y K \frac{\partial}{\partial x} b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,y^3 K \frac{\partial}{\partial x} b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x^4 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x^2 y^2 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,y^4 \frac{\partial^2}{\partial x \partial y} b_x(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^2 y K \frac{\partial}{\partial y} b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
& + \frac{y^3 K \frac{\partial}{\partial y} b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^4 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^2 y^2 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{y^4 \frac{\partial^2}{(\partial y)^2} b_x(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x^3 K \frac{\partial}{\partial x} b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x y^2 K \frac{\partial}{\partial x} b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,x^4 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
& + \frac{i\,x^2 y^2 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{i\,y^4 \frac{\partial^2}{(\partial x)^2} b_y(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^4 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x^2 y^2 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{y^4 \frac{\partial^2}{\partial x \partial y} b_y(u,x,y)}{8\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{x^3 K \frac{\partial}{\partial y} b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} \\
& - \frac{x y^2 K \frac{\partial}{\partial y} b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{5\,x^2 K b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{i\,x y K b_x(u,x,y)}{2\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{3\,y^2 K b_x(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{3i\,x^2 K b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \frac{x y K b_y(u,x,y)}{2\left(x^2+y^2+1\right)^{\frac{3}{2}}} - \frac{5i\,y^2 K b_y(u,x,y)}{4\left(x^2+y^2+1\right)^{\frac{3}{2}}} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_4 = & -\frac{x^8 K^3 \frac{\partial^2}{(\partial x)^2} H}{4\,(x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1)} \\
& - \frac{x^6y^2K^3 \frac{\partial^2}{(\partial x)^2} H}{x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1} \\
& - \frac{3\,x^4y^4K^3 \frac{\partial^2}{(\partial x)^2} H}{2\,(x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1)} \\
& - \frac{x^2y^6K^3 \frac{\partial^2}{(\partial x)^2} H}{x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1} \\
& - \frac{y^8K^3 \frac{\partial^2}{(\partial x)^2} H}{4\,(x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1)} \\
& + \frac{i\,x^8K^3 \frac{\partial^2}{\partial x \partial y} H}{2\,(x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1)} \\
& + \frac{2i\,x^6y^2K^3 \frac{\partial^2}{\partial x \partial y} H}{x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1} \\
& + \frac{3i\,x^4y^4K^3 \frac{\partial^2}{\partial x \partial y} H}{x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1} \\
& + \frac{2i\,x^2y^6K^3 \frac{\partial^2}{\partial x \partial y} H}{x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1} \\
& + \frac{i\,y^8K^3 \frac{\partial^2}{\partial x \partial y} H}{2\,(x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1)} \\
& + \frac{x^8K^3 \frac{\partial^2}{(\partial y)^2} H}{4\,(x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1)} \\
& + \frac{x^6y^2K^3 \frac{\partial^2}{(\partial y)^2} H}{x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1} \\
& + \frac{3\,x^4y^4K^3 \frac{\partial^2}{(\partial y)^2} H}{2\,(x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1)} \\
& + \frac{x^2y^6K^3 \frac{\partial^2}{(\partial y)^2} H}{x^6K + 3\,x^4y^2K + 3\,x^2y^4K + y^6K + x^6 + 3\,x^4y^2 + 3\,x^2y^4 + y^6 + 3\,x^4K + 6\,x^2y^2K + 3\,y^4K + 3\,x^4 + 6\,x^2y^2 + 3\,y^4 + 3\,x^2K + 3\,y^2K + 3\,x^2 + 3\,y^2 + K + 1} + \dots
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"